

■ Video Analysis & Automated Forensic Crawling Platform

■ Comprehensive Manager & Technical User Guide

Executive Summary

The **Video Analysis & Automated Forensic Crawling Platform** is an enterprise-grade digital forensics system designed to investigate, analyze, and document illegal online gambling, betting platforms, and financial fraud sites.

The platform combines **Playwright-stealth web crawling**, **automated 1080p screen recording**, **OpenCV computer vision & OCR**, **Google Gemini Vision LLM screenshot summarization**, and a **Streamlit analytical dashboard** with RESTful FastAPI endpoints.

■■ System Architecture & Core Stack

Core Technologies Used:

- **Backend API:** FastAPI, Uvicorn, SQLAlchemy, PostgreSQL
- **Frontend Dashboard:** Streamlit, Plotly, HTML5/CSS3 Custom Components
- **Web Automation:** Python Async Playwright, `playwright\_stealth`
- **Computer Vision & AI:** OpenCV, Tesseract OCR, PyZBar (QR Codes), Google Gemini 3.5 Vision LLM (`google-gemai`)
- **Reporting Engine:** ReportLab (PDF), `python-docx` (Microsoft Word), Custom HTML5 Base64 Exporter

■ Prerequisites & Installation Guide

1. System Requirements

- **OS:** Linux (Ubuntu 22.04+ recommended), macOS, or Windows WSL2
- **Python:** Version 3.10 to 3.12
- **Database:** PostgreSQL (Running on `localhost:5432` or remote)
- **OCR Engine:** Tesseract OCR (`sudo apt install tesseract-ocr`)

2. Environment Setup

Clone the repository and set up the Python virtual environment:

3. Setting Up Google Gemini API Key

To enable AI-powered screenshot summaries, export your Gemini API key in your terminal session:

(Note: If no API key is provided, the platform automatically uses rule-based fallback summaries so execution never breaks.)

■ How to Run the Platform

Launch the two core services in separate terminal tabs:

Terminal 1: Launch FastAPI Backend Server

- **REST API Base URL:** `http://localhost:8000`
- **Interactive OpenAPI Documentation:** `http://localhost:8000/docs`

Terminal 2: Launch Streamlit Analytical Dashboard

- **Dashboard Web UI:** `http://localhost:8501`

■ Step-by-Step Usage Workflows

Workflow A: Automated Web Crawling & Recording

1. Open the Streamlit Dashboard at `http://localhost:8501`.
2. In the **Sidebar Navigation**, select "**■ Automated Crawler & Live Capture**".
3. **Enter Target URL:** Input the target website address (e.g. `https://cricway.bet`).
4. **Credentials & OTP Handling:**
 - The crawler automatically matches domain names against `rules/seed_accounts.csv`.
 - If the site requires Two-Factor Authentication (OTP), the crawler will pause asynchronously and prompt the user in the Streamlit UI to enter the OTP code received on SMS/WhatsApp.
5. Click "**■ Start Automated Crawl & Video Recording**".
6. The Playwright engine executes the 4-phase forensic crawl:
 - **Phase 1: Authentication & Login**
 - **Phase 2: Context & Game Lobby Exploration**
 - **Phase 3: Affiliate & MLM Profile Scraping**
 - **Phase 4: Financial Gateway & QR Code Generation**
7. Once finished, the session video is recorded, saved, and automatically sent to the analysis pipeline.

Workflow B: Manual Video Upload & Analysis

1. Go to the dashboard sidebar and choose "**■ Direct Video Upload**".
2. Select any screen recording video file (`.mp4`, `.webm`, `.avi`).
3. Click "**Run Forensic Analysis Pipeline**".
4. The system executes:
 - **Frame Extraction:** Detects significant visual scene changes.
 - **Bounding Box Annotation:** Highlights detected payment terms, deposit CTAs, and UPI QR codes.
 - **AI Vision Analysis:** Google Gemini analyzes key proof frames to explain UI components.
 - **Risk Scoring:** Computes Gambling Risk, Banking Risk, and Cryptocurrency Risk percentages.

Workflow C: Reviewing Analytics & Exporting Evidence

Navigate through the 5 Interactive Analytics Tabs:

1. **■ Executive Intelligence Summary:**
 - High-level risk gauge meters (Gambling Score, Banking Score, Crypto Score).
 - Video metadata (duration, frame count, resolution, FPS).
2. **■ Segment Explorer:**
 - Frame-by-frame breakdown with annotated bounding boxes.
 - Displays **■ AI Forensic Screenshot Summaries** explaining each screen.
3. **■ Transaction Funnel & Risk Attribution:**
 - Detailed breakdown of detected payment channels (UPI, Paytm, PhonePe, Bank Transfer, Crypto).
 - Beneficiary account numbers, IFSC codes, and UPI IDs extracted via OCR.
4. **■ Sankey Flow & Gantt Timeline:**
 - Visual flow diagrams mapping user journey from Landing Page -> Game Lobby -> Cashier -> Deposit Execution.
5. **■ Forensic Audit & Report Exports:**

- Export court-admissible evidence packages in 4 formats:
- ■ **PDF Report** (`.pdf`): Full visual report with AI summaries under proof frames.
- ■ **Word Document** (`.docx`): Editable Microsoft Word report formatted for law enforcement submission.
- ■ **HTML Web Report** (`.html`): Interactive standalone web report with embedded base64 images.
- ■ **JSON Data Package** (`.json`): Complete raw structured dataset.

■ API Integration Quick Reference

For developers and automated integrations, all features are exposed via REST API endpoints:

Endpoint	Method	Description
/:---	:---	:---
/`POST /crawl`	`POST`	Triggers 4-phase automated Playwright web crawl and video capture
/`POST /analyses/upload`	`POST`	Uploads a video file for full forensic processing
/`GET /analyses`	`GET`	Lists all historical analysis tasks
/`GET /analyses/{task_id}/summary`	`GET`	Fetches complete analysis metrics, scores, and segments
/`GET /analyses/{task_id}/export/{format}`	`GET`	Downloads report in `pdf`, `docx`, `html`, or `json` format

Interactive API documentation with request/response testing is available at ``http://localhost:8000/docs``.

■■ Maintenance & Seed Configuration

Managing Seed Account Credentials

To store credentials for automated site logins, update ``rules/seed_accounts.csv``:

■ Support & Verification Checklist

Before presenting to management, verify the following:

- [x] PostgreSQL database is running (``sudo service postgresql status``)
- [x] FastAPI server running on port 8000 (``http://localhost:8000/docs``)
- [x] Streamlit UI running on port 8501 (``http://localhost:8501``)
- [x] ``GEMINI_API_KEY`` exported in terminal environment
- [x] Test report exports generated (`` .pdf`, ` .docx`, ` .html`, ` .json``)